

Stage 11E-GR0 Common Grid Geometry and Numerical Diagnostics

mdstats

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Purpose

Stage 11E-GR0 extracts the backend-neutral numerical contracts that were first validated through atomic and framework density plotting. It does not choose a resource-feasible grid ladder, construct a density field, extract an isosurface, or apply a browser policy. Those responsibilities remain with GR1 and later stages.

The analysis-owned layer provides one common Euclidean interpretation of an oblique periodic cell for dense, local-sparse, plotting, and Stage 11 scientific consumers.

Runtime ownership

mdstats/analysis/density/numerical_errors.py

- DensityNumericalError
- DensityNumericalInputError
- DensityNumericalResourceError
- DensityNumericalSerializationError

mdstats/analysis/density/grid_geometry.py

- DensityGridGeometry
- resolve_density_grid_shape
- density_grid_intervals
- prepare_density_grid_geometry
- density_resolution_ratio

mdstats/analysis/density/diagnostics.py

- PeriodicMeanPolicy
- CellEquivalenceReport
- ReciprocalResolutionDiagnostic
- PeriodicMeanDiagnostic

```

PeriodicSpreadDiagnostics
evaluate_cell_equivalence
require_equivalent_laboratory_density_cells
reciprocal_resolution_diagnostic
periodic_frechet_mean_diagnostic
periodic_item_spread_diagnostics

mdstats/analysis/density/stencil_diagnostics.py
    PeriodicGaussianStencilMoments
    gaussian_cutoff_radius
    periodic_gaussian_stencil_moments

mdstats/analysis/density/broadening.py
    ArtificialBroadeningDiagnostic
    cic_assignment_covariance
    effective_artificial_broadening

```

The plotting modules retain compatibility imports and translate analysis-domain input/resource failures into graph-facing exceptions only at the plotting boundary.

Cell and grid geometry

Cell vectors are row vectors. For target Cartesian interval $\mathbf{h}$, automatic grid shape is

$$N_i = \max(4, \lceil \|\mathbf{a}_i\|/h - 10^{-12} \rceil).$$

The realized edge intervals are

$$h_i = \|\mathbf{a}_i\|/N_i.$$

No orthogonalization is permitted. The exact oblique cell matrix, determinant, metric, grid-step vectors, and requested-versus-realized intervals are retained by **DensityGridGeometry**. Explicit grid shapes are validated as three positive integers and are never silently rescaled.

Periodic means and spread

The existing deterministic multi-start flat-torus Frechet/Karcher iteration is the normative implementation. Its start policy, medoid search, ambiguity test, valid-reference mask, weighted spread quantile, temporal subsampling, and metadata schemas remain numerically unchanged.

A periodic mean that is nonconverged or multiply optimal remains a diagnostic result but cannot become a valid adaptive-resolution reference.

Reciprocal resolution

The reciprocal diagnostic is derived from the sampling lattice

$$2\pi \text{diag}(N_1, N_2, N_3) H^{-T}.$$

It records the shortest nonzero reciprocal sampling vector, the deterministic integer representative, and the associated real-space reciprocal interval. The search and tie-breaking rules are unchanged from the tested plotting oracle.

CIC and Gaussian-stencil broadening

For weighted samples, periodic cloud-in-cell assignment contributes a Cartesian covariance determined by each sample's sub-grid phase. The canonical discrete periodized Gaussian stencil contributes its own Cartesian covariance. The analysis layer records

```
C_CIC
C_stencil
C_effective = C_CIC + C_stencil
rms = sqrt(trace(C) / 3)
```

The stencil-moment calculation does not allocate the dense logical stencil. It uses deterministic lexicographic periodic-image enumeration, the existing Gaussian tail tolerance, exact discrete normalization, and the same covariance accumulation order as the plotting oracle.

Exception boundary

The common layer raises only analysis-domain exceptions. It imports no Plotly, mesh, browser, graph-style, graph-complexity, or HTML policy. Plotting adapters translate:

```
DensityNumericalInputError    -> GraphAdapterError / GraphStyleError
DensityNumericalResourceError -> GraphComplexityError
```

Graph-facing exception names are therefore adapter behavior, not common scientific numerical types.

Compatibility contract

The following plotting imports remain available and return the same record types and serialized metadata as their analysis counterparts:

```
mdstats.plotting.density_diagnostics
mdstats.plotting.density_broadening
mdstats.plotting.density_kernel.PeriodicGaussianStencilMoments
mdstats.plotting.density_kernel.periodic_gaussian_stencil_moments
mdstats.plotting.atomic_density.resolve_density_grid_shape
mdstats.plotting.atomic_density.density_grid_intervals
```

For valid inputs, numerical arrays, scalar diagnostics, tuple ordering, schema versions, and metadata dictionaries must be exactly equal or bitwise equal where the previous implementation was deterministic.

Out of scope

GR0 does not own:

- resource-derived target-shape or finest-feasible planning;
- nested-grid ladders or convergence certificates;
- dense/local-sparse backend selection;
- density deposition, convolution, or field normalization;
- marching cubes, mesh repair, Plotly traces, browser admission, or HTML output;
- scientific promotion of a budget-limited grid.

GR1 planning is implemented in 0.20.24a0, and GR2 plotting adaptation is implemented in 0.20.25a0; fixed-kernel convergence and later field-ownership migration remain GR3–GR5 responsibilities.

Acceptance tests

The permanent GR0 boundary covers:

- orthogonal and strongly triclinic automatic and explicit grids;
- exact requested/realized interval serialization;
- analysis-versus-plotting class identity and metadata parity;
- periodic translation invariance of Frechet means and spreads;
- reciprocal-resolution agreement with exhaustive skew-cell enumeration;
- CIC phase covariance in oblique cells;
- stencil-moment parity with the dense canonical stencil;
- effective covariance additivity and RMS consistency;
- immutable diagnostic arrays and metadata;
- import isolation from Plotly, mesh, browser, and graph policy;
- plotting translation of analysis resource failures;

- existing atomic/framework density regression tests.

Implementation status

Implemented in 0.20.23a0. Stage 11E-GR1 is implemented in 0.20.24a0, and Stage 11E-GR2 is implemented in 0.20.25a0. Stage 11E-GR3 is the next implementation stage.